Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad Prov Original

Kurskod	Provkod	Tentamensdatum
MA080G	T102	2025-01-11
Kursnamn	Matematik GR (B), Kryptografi	
Provnamn	Skriftlig tentamen	
Ort	Sundsvall	
Termin		
Ämne		

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Tentamen
MA080G Kryptografi
11 January 2025

Mittuniversitetet
IMD
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Skrivtid: 5 timmar

Hjälpmedel: Miniräknare som ej är symbolhanterande. Ange märke och modell på din miniräknare på omslaget till tentamen. Inga böcker, anteckningar eller mobiltelefoner är tillåtna i tentamensrummet. This is a closed book examination. No books, notes or mobile telephones are allowed in the examination room. Electronic calculators may be used provided they cannot handle formulas. The make and model used must be specifed on the cover of your script.

Till alla uppgifter skall fullständiga lösningar lämnas. Resonemang och uträkningar får inte vara så knapphändiga att de blir svåra att följa. En uppgift per blad, skriv endast på en sida.

Betyg sätts efter hur väl lärandemålen är uppfyllda. Riktvärden för betygen är A 22p, B 18p, C 14p, D 10p och E 9p.

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1. Answer the following questions. **(4p)**
   (a) Encrypt the message
   `"life is fun"`
   using a Caesar cipher with key $k = 25$.
   (b) The message below is a ciphertext encrypted using a Vigenère cipher. Use the Friedman's attack to estimate the length of the key. Is this a good method?

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   ABCDE ABCDE ABCDE ABCDE ABCDE ABCDE ABCDE ABCDE ABCDE
   ABCDE ABCDE ABCDE ABCDE ABCDE ABCDE ABCDE ABCDE ABCDE
   ```

   Hint: Recall that
   $$k \approx \frac{I_E - I_R}{I_C - I_R},$$
   where $I_E \approx 0.0656$, $I_R = \frac{1}{26}$ and the Friedman's coincidence index is
   $$I_C = \left(\frac{n_a}{N} \times \frac{n_a - 1}{N - 1}\right) + \cdots + \left(\frac{n_z}{N} \times \frac{n_z - 1}{N - 1}\right).$$

2. Solve the following problems. **(4p)**
   (a) Recall that an element $x \in G$ is a n-torsion element of the group G if $x^n = e$, where e denotes the identity element. What are the 3-torsion elements of the group $\mathbb{F}_6^\times$?
   (b) Define the Euler's totient function and calculate $\phi(7)$, without using any formula.

3. Use Fermat's little theorem to show that the order of any element in $\mathbb{F}_{23}^\times$ divides 22. **(4p)**

4. Use the Pollard ρ algorithm to factorize $n = 2023$. You must choose the polynomial function $f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ given by

$$f(x) = 2x^2 + x - 2 \pmod{n}.$$

(4p)

5. Bob wishes to send Alice a message encrypted with ElGamal encryption. They decide to use p = 83 and g = 6. Alice picks an integer a in $\mathbb{Z}/83\mathbb{Z}$ and computes $g^a \equiv 50 \pmod{p}$. Bob chooses $b = 15$ and wishes to send the message $m = 82$. **(4p)**
   (a) What is the ciphertext that Bob sends?
   (b) What is Alice's private key a?

6. Alice and Bob use RSA with the same $n = 87$. Their encryption exponents are $e = 3$ and $f = 5$. Charles sends the same message m to Alice and Bob. Somehow, you intercept both of his messages: he is sending the encrypted message $c_A = 6$ to Alice, and the encrypted message $c_B = 9$ to Bob. Is it possible to find out what the plaintext message m is? **(4p)**